

10) 12-bit Hamming coded data including 8-bit actual data and 4-bit parity is received as given below.

Bit	1	2	3	4	5	6	7	8	9	10	11	12
Data	0	0	1	1	1	1	0	1	0	0	0	0

Using the error checking method provided below, detect position of the error (if any) within the code.

C1 = XOR of bits (1, 3, 5, 7, 9, 11) =

C2 = XOR of bits (2, 3, 6, 7, 10, 11) =

C4 = XOR of bits (4, 5, 6, 7, 12) =

C8 = XOR of bits (8, 9, 10, 11, 12)

A) Bit-2

B) Bit-4

C) Bit-6

D) Bit-10

E) No error

11) What is the corresponding value of -76.5 in nearest even rounding mode?

A) -77

B) -75

C) -78

D) -70

E) -76

12) The code below computes the sum of numbers from 1 to N using the formulation below. Then, which instruction should be inserted into the missing line within the code?

$$\sum_{x=1}^N x = N.(N+1)/2$$

```
ORG 100h
CALL sum
RET
```

```
sum PROC
MOV AL, N
MOV BL, N
INC BL
MUL BL
missing instruction?
RET
sum ENDP
```

N DB 8

A) DEC AL

B) SHR AL, 1

C) SUB AL, 2

D) ADD AL, 2

E) SHL AL, 1

13) What is the decimal equivalent of IEEE 754 single precision floating point number C2208000h?

A) -20.25

B) -40.125

C) -10.50

D) NaN

E) +∞

14) What is the size of Personal Computer I/O map?

A) 64K

B) 32K

C) 4K

D) 16K

E) 8K

13) Which flag is tested by a conditional Interrupt instruction?
 A) Carry B) Interrupt C) Overflow D) Parity E) Zero

14) A near jump allows a program to branch forward _____ bytes or backwards _____ bytes from the next instruction's address in a program within the current code segment.
 A) 64K, 64K B) 127, 128 C) 128, 127 D) 255, 256 E) 32K, 32K

15) Which of the followings are interrupt pins found on Intel microprocessors?

- I. INTA
- II. INTC
- III. INTQ
- IV. INTR

A) Only III B) I and IV C) I, II, III D) I, III, IV E) II and III

16) 12-bit Hamming coded data including 8-bit actual data and 4-bit parity is received as given below.

Bit	1	2	3	4	5	6	7	8	9	10	11	12
Data	0	0	1	1	1	0	0	1	0	1	1	0

Using the error checking method provided below, detect position of the error (if any) within the code.

$$C1 = \text{XOR of bits (1, 3, 5, 7, 9, 11)} = 0 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 1 = 0$$

$$C2 = \text{XOR of bits (2, 3, 6, 7, 10, 11)} = 0 \oplus 1 \oplus 0 \oplus 0 \oplus 1 \oplus 1 = 1$$

$$C4 = \text{XOR of bits (4, 5, 6, 7, 12)} = 1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 \oplus 0 = 0$$

$$C8 = \text{XOR of bits (8, 9, 10, 11, 12)} = 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 0 = 1$$

E) Bit-5

C) No error

B) Bit-12

D) Bit-11

A) Bit-3

17) Right shift by 3-bits is equivalent to _____.

- A) divide by 3
- B) multiply by 4
- C) multiply by 3
- D) multiply by 8
- E) divide by 8

18) INC instruction is a special type of addition. However, _____ flag is not affected by INC instruction unlike ADD.

A) P B) Z C) A D) C E) S

?

g) Assume that $DS = 1000H$, $CS = 2000H$, $SS = 3000H$, and $IP = 2000H$. Then, what is the linear address of the instruction fetched by the microprocessor?

D) $22000H$ E) $5000H$

- Assume that $DS = 1000H$, $CS = 2000H$, $SS = 3000H$, and $IP = 4000H$. Which instruction executed by the microprocessor?
- A) $4000H$ B) $12000H$ C) $2000H$ D) $22000H$ E) $5000H$

7) Which of the following statements is not true when an interrupt occurs?

- I. Flags register contents are pushed on the stack.
- II. Interrupt flag is set.
- III. Contents of CS and IP are pushed onto the stack.
- IV. Related ISR is executed.

- ☒ C) I and IV

8) Which of the following communication architectures are serial?

- I. USB
II. PCI
III. Fiber Channel
IV. ISA

5. VISA
- A) I, II, III
- B) I and III
- C) Only I
- D) I, II, IV
- E) I and II

serial communication is superior to parallel communication.

- ☒ A) Always
☐ B) Never
☐ C) Below a particular speed and distance threshold
☐ D) If the amount of data is few
☐ E) Above a particular speed and distance threshold

10) Which of the following conditional jump instructions may follow the comparison of unsigned numbers?

- I. JL
II. JB
III. JE
IV. JZ

- A) II and IV
B) I and IV
C) II and III
D) Only II

11) What is the decimal equivalent of IEEE 754 single precision floating point number C1A20000h?

- What is the decimal equivalent of the following number?
- A) +40.00
- B) NaN
- C) -20.25
- D) -10.50
- E) +8

12) Assume that SP=0100H. What is the value of SP after execution of PUSH AX instruction?

- A) 00FEH
B) 00FFH
C) 0101H

$$131 - 227 = 4$$

ff. 0100

01000191

E) CLC

D) INC

19) Which of the following instructions does not modify the carry flag bit?

A) SUB

B) ADD

C) CMP

I. DS
II. ES
III. CS
IV. SS

20) Which of the following segment registers can not be retrieved from the stack?

A) II, III, IV

B) I and II

C) I, II, IV

D) III and IV

E) Only I

D) III and IV

21) Where is the port number stored for an I/O instruction if the port number is greater than 255?

A) AX

B) AL

C) Variable

D) CX

E) DI

22) It is given that the following code computes factorial of var_N. Then, what is the missing instruction code?

```
MOV AX, 1  
MOV BX, var_N  
MOV CX, var_N
```

label:

MUL BX

missing instruction?

LOOP label

RET

var_N DW 6

A) DEC CX

B) INC AX

C) DEC BX

D) INC BX

23) Which of the following instructions converts ASCII value available in AL register to BCD? Hint. ASCII equivalent of a BCD code can be obtained by appending $(011)_2$ to left.

A) AND AL, 0Fh

B) OR AL, 07h

C) XOR AL, 0Fh

D) AND AL, 07h

E) OR AL, 0Fh

24) What is the size of Personal Computer I/O map?

A) 64K

B) 4K

C) 32K

Group A

- Books, notes, dictionaries, calculators, and cell phones are not allowed.

- You have 60 minutes to solve 25 questions.

- Each question is +4 points.

- Please obey to ethical rules and remember that cheating would result in immediate failure of the course.

- Write your all answers to the answer sheet provided at the last page.

- Good luck.

ID

Name

Grade

1) Instruction Queue is placed in _____ of Intel 8086.

- ☒ A) Control and Timing Unit
- ☐ B) Execution Unit
- ☐ C) Register File
- ☐ D) Bus Interface Unit
- ☐ E) Arithmetic Logic Unit

2) _____ is the act of synchronizing two systems that operate asynchronously.

- ☐ A) Serial communication
- ☐ B) Handshaking
- ☒ C) Transmission
- ☐ D) Parallel communication
- ☐ E) Peripheral interface

3) Which of the followings are interrupt pins found on Intel microprocessors?

- ☒ I. INTA
- ☒ II. INTC
- ☒ III. INTQ
- ☒ IV. INTR

A) I and IV

☒ B) I, III, IV

C) Only III

D) II and

15) How many bits are used for mantissa in IEEE-754 double precision floating point representation?

E) 23

D) 55

C) 24

instruction, and

A) 52

B) 8

16) In 8086, 32-bit addition can be performed by adding low 16-bits of the operands using _____ instruction.
high 16-bits of the operands using _____ instruction.

- A) ADC - SBB
- B) ADD - ADC
- C) ADC - ADC
- D) ADD - ADD
- E) ADC - ADD

17) What is the IEEE 754 floating point equivalent of $-\infty$?

- A) 1000000011111111111111111111111b
- B) 1000000000000000000000000000000b
- C) 0000000000000000000000000000000b
- D) 1111111100000000000000000000000b
- E) 1111111111111111111111111111111b

18) It is given that the following code computes factorial of var_N. Then, what is the missing instruction in the code?

```
MOV AX, 1
MOV BX, var_N
MOV CX, var_N
label:
MUL BX
missing instruction?
LOOP label
RET
var_N DW 6
```

A) INC AX

B) DEC CX

C) INC BX

D) DEC AX

E) DEC BX

19) Which field of an instruction specifies the addressing mode to be used?

A) REG

B) R/M

C) Direction

D) Opcode

E) MOD

20) Assume that CS=1000H, DS=2000H, SS=3000H, and IP=2000H. Then, what is the linear address of the next instruction executed by the microprocessor?

A) 12000H

B) 22000H

C) 1000H

D) 32000H

E) 21000H

Q. 6) Which of the following is correct?

- I. USB
- II. PCI
- III. PCI Express
- IV. SATA

D) Only I

C) I and III

E) Zero

B) I, II, III

A) I, II, IV

D) Overflow

Q. 7) Which flag is tested by a conditional interrupt instruction?

C) Parity

B) Carry

Q. 8) Which of the following statements are true?

A) Interrupt

Q. 9) Regarding SRAM and DRAM, which of the following statements are true?

I. DRAM is faster than SRAM.

II. SRAM is more expensive than DRAM.

III. DRAM requires periodical refresh.

IV. SRAM consumes more power than DRAM.

D) I and IV

E) III and IV

B) I, II, IV

A) II and III

Q. 10) What is the IEEE 754 single precision floating point representation of decimal +10.0?

D) C1200000h

E) C0A00000h

B) C1A00000h

C) 41A00000h

A) 41200000h

Q. 11) Which of the following statements is not true when an interrupt occurs?

I. Flags register contents are pushed on the stack.

II. Interrupt flag is set.

III. Contents of CS and IP are pushed onto the stack.

IV. Related ISR is executed.

D) II and III

E) I and IV

C) III and IV

B) I and III

A) Only II

Q. 12) Using little endian format, 4321h is to be stored in an 8-bit memory starting at 200h. What is the content of memory address 201h after storing?

A) 34h

B) 43h

C) 12h

D) Empty

E) 21h

floating point representation?
D) 55

E) 23

using _____ instruction, and

Which of the following instructions converts any BCD value available in AX register into ASCII format?
Hint: ASCII equivalent of a BCD code can be obtained by appending (011)₂ to left of the BCD code.

- A) AND AX, 07h
- B) OR AX, 0Fh
- C) AND AX, 0Fh
- D) XOR AX, 0Fh
- E) OR AX, 30h

22) Assume that SP=0100H. What is the value of SP after execution of POP AX instruction?

- A) 00FFH
- B) 0101H
- C) 0102H
- D) 00FEH
- E) 0100H

23) Which of the following communication architectures are parallel?

- I. USB
- II. PCI
- III. PCI Express
- IV. ISA

- A) I and II
- B) I, II, III
- C) I and III
- D) II and IV
- E) II and III

24) How many (128Mx16) memory device is needed to store 1 G bytes of data?

- A) 1
- B) 16
- C) 2
- D) 8

$$\begin{array}{r} 128 \\ \times 16 \\ \hline 2048 \\ \hline 128 \end{array}$$

25) Which of the following connections are common to all memory devices?

- I. Address
- II. Read Control
- III. Write Control
- IV. Data

- A) I, II, IV
- B) I, II, III, IV
- C) Only IV
- D) I, II, III

E) II